FOR A DERIVED COMPLETE OBJECT THE VANISHING ABOVE DEGREE ZERO CAN BE CHECKED AFTER TENSORING WITH THE KOSZUL COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'We have a distinguished triangles' in the counted proof span; the grammatical slip is reproduced unchanged. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Derived complete objects of D(A) and the fact that a derived complete object on which multiplication by an element is surjective in positive degrees vanishes there. used at proof lines 26461-26468. Is what lets the surjectivity of f_s produce the vanishing in the descending induction.
- The Koszul complex K_bullet(A, f_1, ..., f_s) with its cone description. used at proof lines 26451-26461. Supplies the objects K(s) and the distinguished triangles along which both inductions run.
- The bounded-above-mod-I lemma applied modulo the ideal. invoked at proof line 26450. Is the implication (2) implies (3).

Source reference. Original lemma 24110: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24110–24110. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 6751: sources/stacks-more-algebra/pinned-original-source.tex, source lines 6751–6751. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 26100: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26100–26100. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 26068: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26068–26068. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original definition 26305: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26305–26305. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $I = (f_1, \ldots, f_r)$ be a finitely generated ideal of a ring A. Let K be a derived complete object of $D(A)$. The following are equivalent*

(1) $H^i(K) = 0$ *for* $i > 0$,
(2) $H^i(K \otimes^{\mathbf{L}}_A A/I) = 0$ *for* $i > 0$,
(3) $H^i(K \otimes^{\mathbf{L}}_A K_1^\bullet) = 0$ *for* $i > 0$ *where* $K_1^\bullet$ *is as in Situation 26305.*

Proof. The implication (1) $\Rightarrow$ (2) is always true. The implication (2) $\Rightarrow$ (3) follows from Lemma 24110. Assume (3). For $s = 0, \ldots, r$ consider the complex

$$K(s) = K \otimes^{\mathbf{L}}_A K_\bullet(A, f_1, \ldots, f_s)$$

where $K_\bullet(A, f_1, \ldots, f_s)$ is the Koszul complex placed in cohomological degrees $-s, \ldots, 0$. We have a distinguished triangles

$$K(s-1) \xrightarrow{f_s} K(s-1) \to K(s) \to K(s-1)[1]$$

by Lemma 6751. Since $K(0) = K$ is derived complete, it follows by ascending induction that each $K(s)$ is a derived complete object of $D(A)$. By descending induction we'll show that $K(s)$ has no nonzero cohomology in degrees > 0. Namely, this holds for $s = r$ by our assumption (3). If it holds for $K(s)$ and $s > 0$, then $f_s : H^i(K(s-1)) \to H^i(K(s-1))$ is surjective for $i > 0$ and we conclude that $H^i(K(s-1))$ vanishes from Lemmas 26068 and 26100. $\square$